



Vital Spring Medical Center Individual Defense Preparation Guide

Concepts, scripts, likely questions and rehearsal plan

INS 204 · System Analysis and Design
Group 9 · 200 Level

How to Use This Guide

Every member submits the same collective report but defends the complete work individually. Your goal is not to memorise 27 pages. Your goal is to explain the chain of reasoning from clinic problem to proposed design and to answer follow-up questions consistently with the report.

Preparation sequence

1. Read the cheat sheet until you can reproduce its headings from memory.
2. Practise the 8–10 minute script below with the actual slides.
3. Study the concepts and questions in this guide.
4. Complete two timed mock defenses: one normal, one with random interruptions.
5. Return to the main report for any answer you cannot justify.

The Defense Story

1. **Problem:** paper scheduling and disconnected records cause measurable failures.
2. **Analysis:** stakeholders, scope, requirements and feasibility define the right system.
3. **Models:** DFDs explain processes; the ERD explains persistent data.
4. **Design:** interfaces, schema, architecture and controls implement those requirements.
5. **Delivery:** phased migration, training and testing reduce operational risk.
6. **Measurement:** explicit targets determine whether the project succeeded.

Eight-Minute Presentation Script

Slide 1, 30 seconds. Good day. My name is [name], matriculation number [number], and I am a member of Group 9. Our project is the Vital Spring Medical Center Clinic Appointment System. Vital Spring is treated in this project as a medium-sized Nigerian outpatient facility.

Slide 2, 45 seconds. The clinic currently uses paper folders, handwritten registers and a physical appointment book. This creates double-booked slots, missing or duplicated records, long queues, high no-show rates and reports that must be counted manually. The core issue is that scheduling, patient records and reporting are disconnected.

Slide 3, 40 seconds. Vital Spring centralises patient profiles, practitioner availability, conflict-free appointments, live queue coordination, visit records, reminders and management reports in one role-controlled system. Our measurable targets include zero double-booking, patient search under five seconds, existing-patient booking within two minutes, median waiting time reduced by 20 percent and 99 percent availability during clinic hours.

Slide 4, 40 seconds. The system has eighteen functional and twelve non-functional requirements. Functional requirements state what it does, such as offering only open slots for booking. Non-functional requirements specify qualities such as five-second search performance, HTTPS encryption, accessibility and recovery within four hours.

Slide 5, 55 seconds. The Context DFD treats Vital Spring as one process exchanging data with five external entities: patients, front desk staff, doctors and nurses, management and the messaging service. The Level-0 DFD decomposes it into six processes and six data stores. Process 3.0 Coordinate Appointments is central because it validates the requested slot against practitioner schedules, reserves it atomically, records the booking source and triggers notification.

Slide 6, 60 seconds. The ERD contains ten entities. A distinctive feature is the separation of availability slots from queue entries: an appointment reserves a planned slot while the queue records what happened after physical arrival. Patient and Practitioner connect through Appointment; Department classifies Practitioner; a completed Appointment may produce one Visit Record. User Account actions are recorded in Activity Log. The model reaches Third Normal Form by removing repeating groups, partial dependencies and transitive dependencies.

Slide 7, 50 seconds. The interfaces support role-specific work: a live patient-flow dashboard centred on queue and waiting time, staged patient registration with duplicate detection, dated-slot booking and an attendance and utilisation report. Double-booking is prevented in three layers: the interface shows only open slots, the server revalidates availability inside a locked transaction, and the database enforces unique slot_id on active appointments. If two staff confirm the same slot, only one transaction commits.

Slide 8, 45 seconds. Vital Spring uses a secure client-server architecture with four access channels. All channels pass through one protected web gateway. Application services separate identity, scheduling and queue, visit documentation, reporting and background messaging. Only application services access the database. Security includes HTTPS, strong hashing, role-based access, session expiry, immutable activity logs and encrypted backups. A daily downtime pack supports continuity during connectivity loss.

Slide 9, 40 seconds. Implementation spans approximately 22 weeks across nine phases: confirm requirements, prototype, build, test, prepare data, train, pilot, deploy and review at 22, 30 and 90 days. Critical tests include simultaneous slot booking, role bypass, duplicate detection, reminder retry and backup restoration.

Slide 10, 30 seconds. The estimated investment is 7.30 million naira, with annual recurring cost of 1.40 million and quantified annual benefits of 4.50 million. The estimated payback is 2.35 years with a five-year benefit-cost ratio of approximately 3.2. These are classroom estimates, but they support the conclusion that Vital Spring is technically, economically and operationally feasible.

Core Concepts You Must Explain

Requirements

Functional requirements describe services or behaviours: authenticate users, register patients, book appointments and generate reports. **Non-functional requirements** describe quality constraints: security, response time, availability, usability and maintainability.

Why acceptance criteria matter: they turn vague wishes into testable conditions. "Easy booking" is vague; "booking completes in no more than three steps and confirmation arrives within two minutes" can be verified.

DFD Versus ERD

DFD	ERD
Models movement and transformation of data. Contains external entities, processes, flows and stores.	Models persistent data structure. Contains entities, attributes, keys, relationships, cardinality and optionality.
Answers: where does information come from, what happens to it and where is it stored?	Answers: what facts must be stored and how are they related?

Keys and Relationships

A **primary key** uniquely identifies a row. A **foreign key** references another table's primary key and enforces a valid relationship. **Cardinality** states how many instances can relate. **Optionality** states whether participation is mandatory.

An appointment requires exactly one patient and one practitioner. A patient or practitioner may have zero appointments initially and many over time. An appointment has zero or one visit record because cancellation and no-show produce none, while one completed consultation produces one record.

Normalization

Normalization reduces duplicated facts and anomalies. UNF has repeating appointment groups inside a patient row. 1NF makes every value atomic and creates one row per appointment. 2NF moves patient facts that depend only on Hospital Number. 3NF moves practitioner, department and location facts that depend on non-key attributes.

Transactions and Concurrency

A database transaction groups operations so they either commit together or roll back. Vital Spring cannot rely only on the interface's slot display because two users can act on stale information. The database uses SELECT ... FOR

UPDATE to lock the slot row, then verifies the status before inserting the appointment. The unique constraint on slot_id is the authoritative final guard.

Likely Questions and Approved Answers

1. Why did you choose this problem?

It has specific operational failures that can be measured: double-booking, record-retrieval delay, no-shows, queues and lack of reports. A central information system directly addresses those causes.

2. Why is this more than an appointment app?

It covers stakeholders, requirements, patient profiles, practitioner availability, live queue management, clinical visit records, reminders, reports, security, audit, migration, training and review. Appointment management is central, but it operates within a complete clinic information system.

3. What is the difference between Context and Level-0 DFD?

The Context DFD shows Vital Spring as one process and defines external interactions. Level 0 decomposes that process into six major internal processes and introduces six data stores.

4. Why are there no data stores in the Context DFD?

Internal stores are inside the system boundary. A Context DFD intentionally abstracts implementation details and focuses on external entities and the data exchanged with the system.

5. Why did you separate queue entries from appointments?

An appointment reserves a planned future slot, while the queue captures what happens after physical arrival: waiting, vitals, called, in consultation, completed or left. Merging them would lose waiting-time insight and force appointment records to carry transient operational states.

6. What prevents double-booking?

The interface shows only open slots, the server locks and revalidates the slot inside a transaction, and the database enforces unique slot_id on active appointments. Only one transaction can commit for a given slot.

7. What happens during simultaneous booking?

Both users may see the slot as open, but only one transaction can lock and reserve it. The other transaction detects the status change and rolls back, returning available alternatives to the user.

8. Why PostgreSQL?

It is mature, open-source and ACID-compliant, with strong transaction support, foreign keys, unique constraints, row-level locking, roles, backup tools and data types suitable for this design.

9. Why Third Normal Form?

3NF removes repeating groups, partial dependencies and transitive dependencies while preserving a clear transactional structure. Patient, practitioner and department facts are each stored once and referenced by keys.

10. Can a receptionist see diagnoses?

No. Receptionists access demographic and scheduling information only. Clinical content in visit records is limited to authorised clinical roles, and every view is logged in the activity log.

More Panel Questions

11. Why a client-server architecture with a web gateway?

It separates presentation, business rules and data storage. Clients cannot reach the database directly, security rules have one authoritative location, and channels can be maintained or scaled independently.

12. Is Vital Spring legally certified?

No. It is a classroom analysis and design proposal that follows the Nigeria Data Protection Act 2023 and general healthcare privacy principles. Certification would require formal legal, operational and security assessment.

13. How are paper records migrated safely?

Inventory folders, identify duplicates, stage data fields, verify critical fields with a second person, reconcile counts and samples, import in controlled batches and retain paper originals for the approved validation period.

14. What happens during power or internet failure?

Before closing, reception prints the next day's appointment list. During an outage, staff use numbered downtime sheets containing essential identity and appointment fields. After recovery, one person captures entries and a second verifies them. UPS, dual connectivity and encrypted backups reduce impact.

15. Why exclude billing, pharmacy and laboratory modules?

They add separate workflows, integrations and regulatory complexity. The first release is deliberately focused on the urgent scheduling, patient-flow, records, reminders and reporting problems. Additional modules require separate analysis.

16. How were feasibility costs obtained?

They are transparent 2026 classroom estimates based on development, equipment, training, hosting, messaging, support and operational savings. They are not procurement quotations and should be replaced with vendor figures before implementation.

17. How will success be measured?

Against a pre-implementation baseline: double-bookings, search and booking time, no-show rate, median waiting time, uptime, successful backup restoration and user task-completion rates.

18. What is the largest project risk?

A patient-data breach has the highest impact. It is mitigated through least privilege, encryption, strong authentication, immutable activity logs, patching, backups and an incident-response procedure. Migration errors and connectivity loss are also high operational risks.

19. Why do you have ten database entities instead of nine?

Vital Spring separates availability_slot from queue_entry as independent entities. The slot table manages dated, bookable periods, while the queue table tracks physical arrival events and timestamps. This design provides waiting-time data that a combined approach would lose.

20. What would you improve in a real implementation?

Validate assumptions through field observation, perform a data-protection impact assessment, obtain vendor quotations, usability-test with clinic users, implement automated tests and monitoring, and run an independent security review before production.

Rehearsal and Delivery

Three Rehearsals

1. **Accuracy rehearsal:** present without timing; peers stop every unclear or incorrect statement.
2. **Timed rehearsal:** complete all slides in eight minutes, leaving margin for equipment delay.
3. **Hostile rehearsal:** peers interrupt with random questions, ask “why?” twice and request an example.

Presentation Behaviour

- Look at the panel, not continuously at the screen.
- Explain one main idea per slide; never read every label in a diagram.
- Use the cursor only to identify the process or entity being discussed.
- Pause after the question, answer directly, then justify.
- Say “the proposed design” rather than pretending the system has been built and deployed.
- Do not claim legal certification, real patient data or confirmed vendor pricing.

When You Do Not Know

Do not invent. Say: “I do not want to give an inaccurate answer. The design establishes [relevant principle]. The exact implementation detail would be confirmed during development or formal assessment.” Then return to what the report actually specifies.

Final Checklist

Night before	At the venue
Read cheat sheet; rehearse twice; charge devices; keep PPTX and PDF on laptop, phone and USB; know your name/matric slide.	Test projector ratio and fonts; open the PDF backup; close notifications; confirm presentation time; breathe and begin with the problem.

Final principle: defend decisions, not pages. For every design choice, be ready to state the problem it solves, the requirement it satisfies and how it will be tested.